



LOOP LENS

See the physical truth behind your sustainability numbers.

Material Flow Analysis Methodology

How Loop Lens maps material flows and turns them into investor-grade circularity data

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This document describes how Loop Lens performs material flow analysis: the standards it rests on, the procedure we follow, the data we ask clients for, and what an engagement produces. It is written for partners and prospective clients who want to know what is behind the numbers before they rely on them.

1 What a Loop Lens MFA is

Loop Lens is a platform and consultancy turning material flows into measurable value. It offers accessible MFA and circularity baselining to purpose-driven organizations.

MFA maps physical stocks and flows moving through a system over a defined period of time. Whereas an LCA models environmental impact, and sustainability reports disclose specific figures, an MFA serves as the traceable data layer that sits underneath both. It maps what comes in, what goes out, and what accumulates in a closed system.

Central to MFA analysis, the law of mass conservation states that inputs are equal to outputs plus stock change. In other words, mass is neither created nor destroyed in a closed system. MFA responds to three unanswered questions: what is the quantity of material actually moving through the business, where does that material go, and how much material is sitting in stock unnoticed.

Loop Lens does not provide assurance, LCA, or ESG ratings. These are critical functions but fall outside Loop Lens's scope. Rather, by capturing the physical flow of real quantities of materials, Loop Lens's MFA provides the missing layer sitting underneath other layers of sustainability reporting.

2 Standards stack

Loop Lens builds upon a rich canon of sustainability reporting to deliver its rigorous circularity analysis. The following standards and tools lie at the foundation of Loop Lens's methodology.

Brunner and Rechberger

Loop Lens's MFA procedure draws on the canonical text, Brunner and Rechberger's Practical Handbook of Material Flow Analysis. Brunner and Rechberger take the concept of the anthroposphere — the part of the Earth system that has been made, changed, or used by human beings — and establish a rigid, transparent and useful methodology to investigate the material metabolism of such systems. Their text provides the analytical foundation to apply the MFA procedure in order to determine the main sources, flows, stocks, and emissions of man-made and natural materials in anthroposphere systems.

STAN

Loop Lens employs cutting edge software to perform material flow analysis. STAN is a free MFA software from TU Wien that implements the Brunner method. We model MFA in STAN, leveraging the software's data reconciliation, uncertainty propagation, and Sankey export. We consult Claude's frontier models to help structure data collection steps and polish initial drafts of reports. Every figure and conclusion in a Loop Lens report is reviewed and signed off by a named analyst.

The ISO 59000 family

First published by the International Organization for Standardization (ISO) in 2024, the ISO 59000 family of standards sets forth requirements, guidance and tools for organizations to measure and assess their circularity performance within defined economic systems. The three circular economy standards published in 2024 are:

ISO 59004 Terminology, Principles, and Guidance for Implementation

ISO 59010 Guidelines for the Transition of Business Models and Value Networks

ISO 59020 Measurement and Evaluation of Circularity

By standardizing how organizations collect and calculate data, the ISO 59000 family offers universal frameworks for defining, measuring, reporting and enacting circular economy principles in business. Together these standards provide a framework that lets Loop Lens assess, implement, and report on the circular economy at different scales. Loop Lens uses the ISO 59000 frameworks to make robust, measurable, and transparent circular economy recommendations.

Material Circularity Indicator

Introduced by the Ellen MacArthur Foundation in 2015 and updated in 2019 to include materials of biological origin in the assessment scope, the Material Circularity Indicator (MCI) provides a foundation for the quantitative analysis of material cycles and inspired the development of differentiated indicators across industries. The headline indicator gives a value between 0 and 1, where higher values indicate higher circularity.

ESRS and GRI

Loop Lens acknowledges European leadership on circularity and borrows from the EU's rich array of frameworks and theories to deliver insightful circularity analysis. In particular, the European Sustainability Reporting Standards (ESRS), first published in the Official Journal of the European Union in December 2023 and revised in July 2026, establish a comprehensive framework for organizations to report on their transition from a linear "take-make-waste" model to a circular economy. The ESRS are mandatory for companies in scope of the Corporate Sustainability Reporting Directive; the July 2026 revision simplified them substantially and introduced a separate voluntary standard for smaller companies.

ESRS E5-4 (Resource Inflows) requires disclosure of the total weight, in tonnes or kilograms, of all products and materials used across operations and upstream value chains. ESRS E5-5 (Resource Outflows) covers products designed for circularity, including expected durability, reparability, and recycled content. E5-4 and E5-5 fulfill material inflow and waste reporting categories parallel to GRI 301 (Materials) and GRI 306 (Waste). Loop Lens maps client data to the revised standards rather than the 2023 originals.

3 The procedure

The six essential steps for MFA analysis, adapted from the Practical Handbook of Material Flow Analysis, are:

- 1. Define objectives.** State the core questions to be addressed and identify the decisions the analysis supports.
- 2. Define the system.** Loop Lens sets the spatial boundaries — factory, site, or city — and defines the temporal bounds, typically one calendar year. We identify and map relevant goods, materials and indicator elements. Following Brunner and Rechberger, we distinguish goods, which are tradeable items, from substances, which are the elements or compounds they are made of. The question decides which level the analysis runs at.
- 3. Determine material flows.** Identify inputs, outputs, and internal transfers, and sketch a preliminary qualitative process-flow structure.
- 4. Collect and merge data.** Gather data from purchasing, warehouse records, gate weights, literature, or emission factors.
- 5. Calculate mass flows and stocks.** Apply mass balance per process, solving for unknown flows or reconciling balancing discrepancies. Results are visualized using Sankey diagrams or mass-balanced tables.
- 6. Interpret and develop recommendations.** We identify weak points and resource inefficiencies. Then we deliver recommendations for the organization to improve its circularity performance.

Two rules we apply throughout

Every figure in a Loop Lens balance carries a provenance tier, recorded on the line where it appears: measured, invoiced or weighed, inferred from a conversion factor, estimated, or taken from literature. A client should be able to see at a glance which numbers were counted and which were derived.

The second rule concerns what happens when a balance does not close, which is most of the time. There are three honest responses: find the missing flow, adjust within stated uncertainty, or report the difference as an unaccounted residual. Loop Lens does not close a balance by adjusting a number it has not measured. A residual is usually the most useful finding in the exercise, because it points at the part of the operation where material is moving without being recorded.

4 Data requirements

Most organizations arrive at an MFA already carrying two separate reporting burdens. European buyers ask for ESRS-aligned inflow and outflow figures, while national regulators ask for volume declarations under EPR. The same material data serves both obligations if it is collected once, properly.

Loop Lens works with each organization to source the data needed to build a complete analysis and establish its circularity baseline — the table below shows what we ask for and where it typically already lives.

Data category	What to ask for	Likely source system	Acceptable format	Fallback if unavailable
Goods inflow	Quantity and description of every material or product received during the reference year, by weight or unit count	ERP, purchase ledger, goods-received notes	CSV or Excel export, one row per line item	Supplier invoices for the ten largest inputs by spend, extrapolated across the year and labelled as inferred
Goods outflow	Quantity of finished product, resold goods and byproducts leaving the site in the same period	Sales and dispatch records, delivery notes	CSV or Excel export, one row per shipment	Annual sales volumes by product line, converted to mass using product specifications
Stock	Opening and closing inventory for the reference year, in the same units as inflow and outflow	Inventory system, annual stocktake	Two dated snapshots	A single closing stocktake, with the change treated as an unknown and solved within the balance
Composition	Material breakdown of the main goods: what they are made of, and in what proportion	OEM specifications, bills of material, laboratory assay	Spec sheets, BOM export, or assay report	Published composition data for the equivalent product class, labelled as a literature value
End-of-life destination	Tonnage sent to each waste, recycling or recovery route, and the name of each receiving facility	Waste contractor invoices, weighbridge tickets, transfer notes	Twelve months of invoices, or a contractor summary	Contracted collection frequency and container size, converted to mass and labelled as estimated

Waste contractor invoices are the single most underused source in this work. They are weighed, dated and generated by a third party, and they usually sit in accounts payable rather than anywhere near the sustainability team.

Data shared with Loop Lens stays with Loop Lens. We do not publish or share client data without written agreement, and any use in a case study requires the client’s approval.

5 Case study: a medical equipment reconditioner

We performed material flow analysis for a U.S. distributor of medical imaging equipment operating six facilities, covering fiscal year 2025. We are leaving the company anonymous at the request of its private equity investors, but discuss our findings for a business of 280 full-time employees and \$141 million in annual revenue.

As a leading distributor of new and reconditioned medical imaging equipment, the business supported sustainability objectives by extending equipment lifecycles. Material flows were tracked by equipment modality and organized into inflows (equipment and materials entering the company’s operations), internal processing (reconditioning and inventory), and outflows (equipment and materials leaving the company’s operations).

The following metrics indicate the company’s circular economy performance:

Indicator	Value	Interpretation
Component reuse rate	90%	High reuse; substantially reduces virgin material input
Trade-in collection rate	84 units/year	Active take-back program
Reconditioned sales share	6.7% of revenue	Opportunity for growth
Units reconditioned vs. disposed	60:33 (1.8:1)	Positive circularity ratio
Reusable packaging rate	35%	Moderate; improvement opportunity
Recycling rate (excl. equipment)	~16%	Metals and cardboard against total waste

What the numbers mean

The company’s stakeholders asked for reporting to align with ISSB recommendations. The component reuse rate reported above underpins our estimated Material Circularity Indicator of 0.9 for the reconditioning line. An MCI at that level reflects equipment reconditioning as a core competency of the business. It significantly reduces demand for virgin materials and extends the useful life of critical raw materials.

That figure matters because we calculated it from the company’s own operating records rather than from an industry benchmark. Any reconditioner can describe itself as circular; a number derived from its own trade-in, inventory and disposal data is defensible in front of an investor. The caveat belongs alongside it. The 0.9 estimate describes the reconditioning line, which accounted for 6.7% of revenue in FY2025, and not the business as a whole.

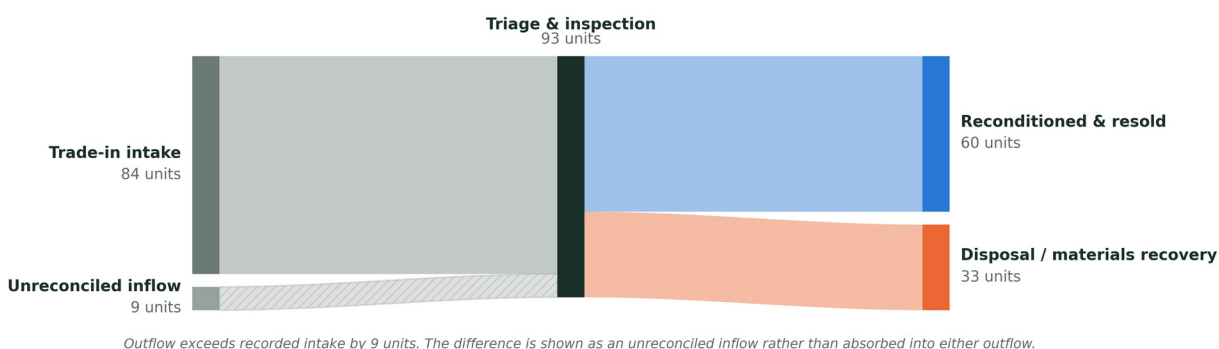
Processes, flows and substances

We built the analysis around six processes: intake from trade-in and purchase, triage and inspection, reconditioning, parts harvesting, sale, and disposal to a licensed recovery facility. Equipment inventory and harvested parts inventory were treated as stocks. The flows we quantified were units in by modality and source, units out by disposition, the change in both inventories, tonnage sent to the recycler, and film recovered through the company's existing recycling program.

Unit counts were converted to mass using published OEM shipping weights, which we record as an inferred value rather than a measured one. Beneath the goods layer we tracked the substances that carry the most value and the most risk: copper, steel and aluminium in the structural components; neodymium-iron-boron in permanent-magnet systems; niobium-titanium and liquid helium in superconducting magnets; and printed circuit boards carrying gold and silver. Rare earths and helium are where a reconditioner's circularity argument is strongest, because a system that stays in service for another decade is a system whose critical raw materials have not been dispersed.

Figure 1 • Unit flows, FY2025

Medical equipment reconditioner (anonymized). Flows are counts of imaging systems, not mass.



The diagram is deliberately unbalanced. Ninety-three units left triage against 84 units of recorded trade-in intake. Rather than absorb the nine-unit difference into one of the outflows, we show it as an unreconciled inflow and report it as an open item. This is the reconciliation rule in Section 3 applied to a real balance.

6 Deliverable specification

Engagements with Loop Lens result in a rigorous set of analyses that can enhance a company's circularity performance.

Once an LOI has been signed establishing the boundaries and timeline for a partnership, Loop Lens will send the client an MFA data collection spreadsheet. The spreadsheet provides detailed instructions for completing data collection, and includes the following tabs: organization information, material and waste, ESRS E5 or circularity metrics, and energy and emissions.

The engagement follows a four-week structure: week 1 for scoping and data requests, week 2 for collection and chasing, week 3 for modeling and reconciliation, and week 4 for drafting and review. The schedule assumes the client returns the completed data collection spreadsheet by the end of week 2; where data arrives later, the timeline moves with it.

Loop Lens deliverables are adjusted to meet client needs, but generally include a report with a Sankey diagram, a balance table with a provenance tier per line, an indicator set, a disclosure mapping, and a prioritized data-gap register.

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— Sources and further reading

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